HACKFUSION 2.0 – 24Hours National Level Hackathon

Problem Statements

Theme: Artificial Intellligence

1. Enterprise Knowledge Assistant

Problem. Employees struggle to find the right clause across policies, SOPs and manuals, often read outdated versions, and generic AI tools answer confidently even when the documents are silent.

Objective. Build an assistant that answers questions only from a supplied document corpus, cites the exact section used, and says so when the corpus does not contain the answer.

- Ingest PDF, TXT and Markdown documents, keeping document, section and page for every passage.
- Retrieve relevant passages using semantic or hybrid search.
- Generate answers grounded only in retrieved text, with section-level citations.
- Classify each response as answered, partially answered, or not covered by the documents.
- Handle conflicting document versions by preferring the current one.
- Provide a simple upload-and-ask interface and a batch mode that writes answers in a fixed output format.

Input. An organizer-authored corpus for a fictional company (40–60 documents, including superseded versions and tables) and a public question set.

2. Smart Classroom Monitoring and Examination Integrity System

Educational institutions increasingly rely on computer laboratories for teaching, practical examinations, programming assessments, and online evaluations. Instructors and invigilators face difficulties monitoring many student computers simultaneously. Students may access prohibited websites, switch applications, use unauthorized AI tools, or communicate during examinations. Traditional physical invigilation cannot continuously observe every screen, while existing solutions may not provide the customizable examination-focused alerts, reporting, and controls an institution requires.

Problem.

Design and develop a centralized classroom monitoring and management system that allows authorized teachers and examination invigilators to monitor multiple student computers in real time from one dashboard. Lightweight agents installed on institution-managed student computers should securely provide screen previews, application activity, and examination events. The platform must detect potential examination-policy violations, generate timely alerts, support classroom controls, and work efficiently over local networks. Monitoring must be limited to authorized educational sessions with appropriate visibility and safeguards.

Technical Requirements:

Component	Expected Capabilities
Student Agent	Lightweight service for institution-managed lab computers
Instructor Dashboard	Web or desktop grid view and focused individual screen
Communication	Encrypted WebSocket, WebRTC or comparable real-time channel
Monitoring	Screen preview/streaming, active-window information and rule-based alerts
Server	Session handling, device enrollment, event aggregation
Security	Authentication, encryption, audit logs and role permissions
Platform Support	Windows and Linux initially; macOS as an extension
Deployment	LAN operation with optional centralized/cloud administration

3. Krishi Mitra

India has Central and State Government initiatives supporting farmers through income assistance, crop insurance, irrigation support, agricultural equipment, and related benefits. Many farmers struggle to discover and compare suitable schemes because information is dispersed across portals, terminology is complicated, and digital services often require text entry in formal language. The challenge becomes harder when a farmer speaks Kannada in a regional accent, uses local crop names, mixes languages, or does not know the exact technical terms required by an application.

Challenge statement: Develop a voice-first AI assistant that understands farmers speaking naturally in Kannada, gathers their agricultural details through conversation, and recommends potentially suitable government schemes using transparent and verifiable eligibility rules.

Objectives

- Recognize natural Kannada speech, including regional accents and common code-mixed phrases.
- Extract location, land area, crops, ownership, season, and assistance needs.
- Ask meaningful follow-up questions when required information is missing.
- Recommend potentially applicable schemes using a rule-based or otherwise auditable eligibility layer.
- Explain reasons, benefits, conditions, documents, and application steps in spoken Kannada.
- Retain context across follow-up questions in the same conversation.

NOTE The main innovation should be the quality of the voice interaction—not merely adding a microphone to a generic text chatbot.

Reference Scheme Dataset

Scheme	Use in the prototype
PM-KISAN (real)	Income assistance for eligible landholding farmer families; verify official exclusions and enrollment requirements.
PMFBY (real)	Crop insurance subject to crop, notified area, season, and applicable government rules.
Demo Micro-Irrigation Support (fictional)	Synthetic test case for drip irrigation support; no real entitlement should be implied.
Demo Agricultural Equipment Assistance (fictional)	Synthetic test case for equipment subsidy decision rules.
Demo Soil Improvement Support (fictional)	Synthetic test case for additional rule-based recommendation paths.

Theme: Computer Vision

1. PPE Compliance Detection (Computer Vision)

Problem. Supervisors cannot consistently spot workers without helmets or safety vests across busy industrial areas.

Objective. Build a vision system that detects each worker, determines whether that specific worker is wearing the required equipment, and flags violations.

Core requirements

- Detect people, helmets and safety vests in images (short video clips optional).
- Associate each item of equipment with the person wearing it.
- Apply predefined compliance rules per worker (for example, helmet and vest required).
- Flag each non-compliant worker and the missing item.
- Display annotated images with a per-worker verdict.
- Generate a violation summary.
- Handle multiple workers, partial occlusion, distance and poor lighting.

Input. Images, and optionally short clips, of workers in compliant and non-compliant combinations, including hard cases such as a helmet held in hand or placed on a table, and high-visibility clothing that is not a vest.

Constraint. A helmet detected anywhere in the image must not be taken to mean every worker in it is compliant.

2. Computer Vision-Based Counting of Overlapping and Stacked Jute Bags

Warehouses may contain thousands of jute bags filled with grains, rice, wheat, pulses, or other agricultural commodities. Bags are frequently arranged in dense, irregular piles, multi-layer stacks, and partially overlapping configurations. Manual inventory counting is slow and can lead to missed or duplicate counts. The visual problem is difficult because many bags have nearly identical colors and textures, boundaries are unclear, lighting is inconsistent, and some objects are partly or completely hidden behind others.

Problem: Build a computer vision system that detects and counts large quantities of jute bags in realistic warehouse imagery, even when bags are irregularly arranged, stacked, partly obscured, or overlapping.

Objectives

- Accept warehouse photographs and optionally videos.
- Identify distinguishable bag instances in dense scenes.
- Separate partially overlapping instances and handle multiple visible stack layers.
- Aggregate counts across sections without double counting.
- Visualize detections with boxes, masks, or markers.
- Report confidence and highlight low-visibility regions.
- Differentiate directly observed bag counts from any estimate of hidden inventory.

Reference Dataset

Test difficulty	Scene properties
Level 1 — easy	Clearly visible bags with minimal overlap.
Level 2 — moderate	Irregular placement and noticeable partial overlap.
Level 3 — hard	Tightly packed multi-layer stacks and major occlusion.
Level 4 — advanced	Warehouse-scale counting from multi-view images or video.

3. Computer Vision-Based Waste Segregation with Edge Deployment

Improper waste segregation is a major challenge in educational institutions, public places, and residential areas, leading to inefficient recycling, increased landfill waste, environmental pollution, and health hazards. Traditional waste segregation relies heavily on manual sorting, which is time-consuming, labor-intensive, and prone to errors.

Problem: Develop an AI-powered, computer vision-based waste segregation system with edge deployment that can automatically identify and classify waste materials such as plastic, paper, metal, glass, organic waste, and other waste using images captured by a camera. The system should run its trained machine learning or deep learning model directly on an edge device, such as a smartphone, enabling real-time classification with minimal latency and without continuous dependence on cloud connectivity.

Reference dataset:

1. https://github.com/garythung/trashnet
2. https://archive.ics.uci.edu/dataset/908/realwaste

Edge deployment:

Android smartphone: Build a camera-based application using an on-device model.

Theme: Cybersecurity

1. API Security Testing Assistant (Cybersecurity)

Problem. APIs often ship with weaknesses in authentication, authorization, input validation or information handling that are only found after deployment.

Objective. Build a tool that runs authorized security checks against an isolated, intentionally vulnerable API and produces an evidence-based vulnerability report.

Core requirements:

Parse the supplied API specification (OpenAPI preferred) and list endpoints with their expected access rules.

- Run authorized tests using test accounts with different roles.
- Detect selected authentication and authorization weaknesses, including object-level authorization flaws (one user reading another user's data).
- Detect selected input-validation and information-disclosure issues.
- Record request and response evidence for every finding.
- Distinguish vulnerable endpoints from correctly secured ones.
- Recommend a remediation for each finding.

Input. API specification, an isolated vulnerable test API (one resettable instance per team), role-based test accounts, and a written testing scope.

Constraint. Testing is limited to the supplied environment and test accounts. Destructive testing and any traffic to external systems are prohibited.

2. SignalSafe Campus : Recognize, Resist and Report Digital Threats

Educational institutions increasingly rely on digital platforms for academic communication, online payments, learning management systems, social networking, and student services. However, students, faculty, and administrative staff are vulnerable to phishing emails, fraudulent websites, malicious links, QR-code scams, payment fraud, cyberbullying, impersonation, and other digital threats. Many users struggle to identify suspicious digital content before interacting with it, and they may not know how to report an incident or preserve relevant evidence. Delayed reporting can lead to financial losses, compromised accounts, privacy violations, and harm to the campus community.

Problem

Develop "SignalSafe Campus," an intelligent and user-friendly cybersecurity solution that helps campus users recognize, assess, prevent, and report digital threats before they cause harm. The solution should analyze suspicious messages, URLs, QR codes, and reported incidents; provide understandable risk alerts and safety recommendations; and offer a secure mechanism for reporting incidents to authorized campus personnel.

The proposed system should prioritize usability, privacy, accessibility, and rapid response, making cybersecurity awareness and incident reporting accessible to every member of the campus community.

Reference datasets for development

1. https://data.mendeley.com/datasets/hx4m73v2sf/2
2. https://www.unb.ca/cic/datasets/trap4phish2025.html

Important: These datasets do not cover every SignalSafe feature. Cyberbullying reports, payment fraud, and campus-specific incidents will need additional suitable data or carefully designed synthetic examples

3. VerityGrid Campus

Develop “VerityGrid Campus,” a privacy-preserving digital trust platform that enables educational institutions to verify the authenticity of documents and digital content, validate identities, enforce access permissions, and share only the minimum necessary information with authorized recipients.

The solution should provide transparent verification results, explain potential risks, maintain auditable access records, and help campus administrators establish trust in digital interactions without unnecessarily exposing sensitive personal information.

The objective is to demonstrate a practical system that answers three questions before trusting or sharing digital information:

- Genuine: Is the document, identity, or digital content authentic and unaltered?
- Necessary: Is the information being requested essential for the intended purpose?
- Authorized: Is the person or system permitted to access or use the information?

Reference datasets

1. https://huggingface.co/datasets/kriyam-ai/kriyam-tamperflow
2. https://ai.meta.com/datasets/dfdc/